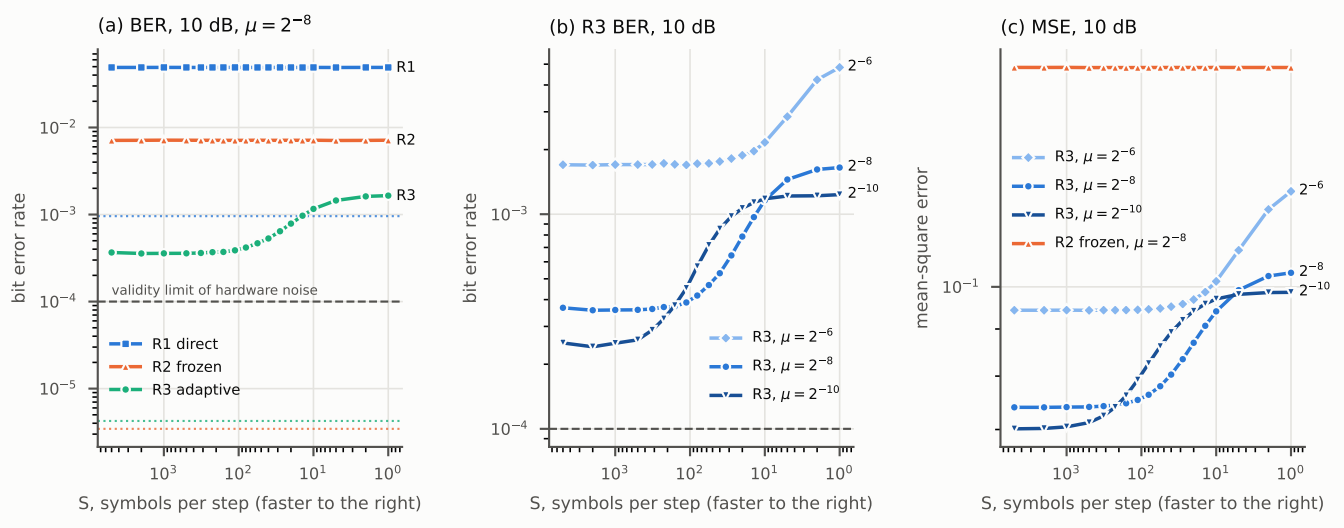
(a) BER, 10 dB, $\mu = 2^{-8}$
bit error rate
10^{-2}
10^{-3}
10^{-4}
10^{-5}
R1
R2
R3
validity limit of hardware noise
R1 direct
R2 frozen
R3 adaptive
10^{3}
10^{2}
10^{1}
10^{0}
S, symbols per step (faster to the right)
(b) R3 BER, 10 dB
bit error rate
10^{-3}
10^{-4}
2^{-6}
2^{-8}
2^{-10}
R3, $\mu = 2^{-6}$
R3, $\mu = 2^{-8}$
R3, $\mu = 2^{-10}$
10^{3}
10^{2}
10^{1}
10^{0}
S, symbols per step (faster to the right)
(c) MSE, 10 dB
mean-square error
10^{-1}
R3, $\mu = 2^{-6}$
R3, $\mu = 2^{-8}$
R3, $\mu = 2^{-10}$
R2 frozen, $\mu = 2^{-8}$
2^{-6}
2^{-8}
2^{-10}
10^{3}
10^{2}
10^{1}
10^{0}
S, symbols per step (faster to the right)